

(10) Seat No : \_\_\_\_\_

No. of printed pages: 03

**SARDAR PATEL UNIVERSITY**  
**B.C.A. (SEMESTER - I) EXAMINATION**  
Thursday, 16<sup>th</sup> Mar. 2017 16/3/17  
2.00 a.m. to 4.00 p.m.  
**MATHEMATICS-I (US01FBCA02)**

**Total Marks: 70**

Q.1 Fill in the blanks:

10

1. Let  $U = \{1, 2, \dots, 10\}$ ,  $A = \{1, 2, 3, 4\}$  and  $B = \{3, 4, 5, 6\}$ . Then  $(A \cap B)^c =$  \_\_\_\_\_  
(a) A (b) B (c)  $\{1, 2, 5, 6, 7, 8, 9, 10\}$  (d) None
2. De Morgan's Law : \_\_\_\_\_  
(a)  $(A \cup B)^c = A^c \cap B^c$  (b)  $(A \cap B)^c = (A \cup B)^c$  (c)  $(A \cup B)^c = A \cap B$  (d) None
3. If  $u = (3, 7, 2)$ ,  $v = (-3, 0, 3)$  then  $u \cdot v =$  \_\_\_\_\_  
(a) -3 (b)  $(6, -7, 0)$  (c) -5 (d) None
4.  $4(1, 2, 1) + 2(1, 3, 3) =$  \_\_\_\_\_  
(a)  $(6, 12, 10)$  (b)  $(6, 10, 14)$  (c)  $(6, 14, 10)$  (d) None
5. A Square matrix A is said to be symmetric if \_\_\_\_\_  
(a)  $A \neq A^T$  (b)  $A = -A^T$  (c)  $A = A^T$  (d) None
6. The inverse of -3 in  $(\mathbb{Z}_7, +_7)$  is \_\_\_\_\_  
(a) 1 (b) 0 (c) 4 (d) None
7. An operation \* on a set S is said to be commutative if  $a * b$  \_\_\_\_\_  $b * a$   
(a) = (b)  $\neq$  (c) < (d) >
8. Mode of 3, 7, 11, 9, 13, 1, 7, 12, 7, 6 is \_\_\_\_\_  
(a) 6 (b) 7 (c) 8.5 (d) None
9. The mean of first 9 natural numbers is \_\_\_\_\_  
(a) 5 (b) 4.5 (c) 5.5 (d) None
10. Geometric mean of x, y, z, w is given by \_\_\_\_\_  
(a)  $\sqrt{xyzw}$  (b)  $\sqrt{x + y + z + w}$  (c)  $\sqrt[4]{xyzw}$  (d) None

Q.2 Attempt any Ten.

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1. If  $A = \{m, f, w, s\}$  and  $B = \{m, t, w, f, th\}$ . Then Find  $A \cup B$  and  $A \cap B$ .
2. Explain: (a) Subset. (b) Intersection of two sets
3. If  $f(x) = x + 5$  and  $g(x) = 3x + 2$  then find  $f \circ g$ .
4. Let  $B = \{0, 1\}$ . Is B closed under: (a) Multiplication? (b) Addition?
5. Define Group with example.
6. In  $\mathbb{Z}_8$  find all roots of  $f(t) = t^2 + 6t$ .

(1)

(P.T.O.)

7. Find  $x$  &  $y$  if  $x(1, 1) + y(2, -1) = (1, 4)$

8. If  $A = \begin{bmatrix} 1 & 2 \\ 7 & -2 \end{bmatrix}$  and  $B = \begin{bmatrix} 4 & -2 \\ 5 & 1 \end{bmatrix}$  then find  $A + B$  and  $A - B$ .

9. Determine the value of  $k$  for which  $\begin{vmatrix} 2k & k \\ 4 & k \end{vmatrix} = 0$ .

10. Find median of the following data;

(1) 8, 12, 20, 40, 8, 7, 5

(2) 7, 7, 8, 6, 5, 8, 3, 2, 2, 9, 7

11. The I.Q. of 10 students is given below:

95, 98, 107, 100, 101, 110, 70, 120, 88, 83. Find the Arithmetic mean of I.Q.

12. Write down various measures of central tendency. Explain one of them with example.

Q.3 (a) Prove that  $1^2 + 2^2 + 3^2 + \dots + n^2 = \frac{n(n+1)(2n+1)}{6}$ . 05

(b) If  $A = \{1, 2, 5, 6, 8\}$ ,  $B = \{x: x \text{ is even}, x \leq 10, x \in N\}$  and  $C = \{1, 2, 3, 5, 6, 11, 12\}$ . Then verify 05

$$1. A \cap (B \cup C) = (A \cap B) \cup (A \cap C)$$

$$2. A \cup (B \cap C) = (A \cup B) \cap (A \cup C)$$

Q.3 OR

(a) Let  $a$  and  $b$  denote positive integers. Suppose a function  $Q$  is defined recursively as: 04

$$Q(a, b) = \begin{cases} 0 & , \text{if } a < b \\ Q(a - b, b) + 1 & , \text{if } b \leq a \end{cases}$$

1. Find the value of  $Q(2, 4)$ ,  $Q(15, 3)$ ,  $Q(17, 4)$ .

2. Find  $Q(2143, 9)$ .

(b) If  $f(x) = x^2 + 5x$  and  $g(x) = 3x + 2$  then find  $gof(x)$ . 03

(c) Find  $5!$  Using recursive definition. 03

Q.4 (a) Consider the group  $G = \{1, 2, 3, 4\}$  under the multiplication modulo 5. Find multiplication table of  $G$ . Also find  $2^{-1}$  and  $4^{-1}$ . 04

(b) Show that  $f: G \rightarrow G'$  defined by  $f(a) = 2^a$  is a homomorphism where  $G$  is a group of real numbers under addition and  $G'$  is a group of positive real numbers under multiplication. 03

(c) Let  $A = \{1, 3, 5, 7, \dots\}$  then check whether  $A$  is closed under 03

1. Multiplication?
2. Addition?

(2)

Q.4

OR

- (a) Consider the set  $N$  of the natural numbers and let  $*$  be the operation of least common multiple (l.c.m) on  $N$ .

04

(1) Find  $4 * 5, 6 * 8, 1 * 8$  and  $7 * 14$ .

(2) Is  $(N, *)$  a semi group? Is it commutative?

- (b) If  $S$  is a nonempty set with the operation  $a * b = a$ .

03

Is the operation: (i) associative? (ii) Commutative?

- (c) In  $Z_{10}$ , find  $-3, -8, 3^{-1}$ .

03

Q.5

- (a) Consider the vectors  $u = (1, -1, 2, -5), v = (-2, 3, -4, 5)$ .  
Find  $u + 3v, 2u - v$  and  $u \cdot v$ . Also find  $\|u\|$  and  $\|v\|$ .

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- (b) Using Cremer's rule solve the simultaneous equations  $5x - 3y = 1, 6x - y = 9$ .

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Q.5

OR

- (a) Define following matrices:

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1. Symmetric matrix

2. Square matrix

3. Orthogonal Matrix.

- (b) If  $A = \begin{bmatrix} 1 & -2 & 3 \\ 6 & 0 & 9 \\ 5 & -7 & 11 \end{bmatrix}$  then find  $A + A^T, A - A^T$  and  $A A^T$ .

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- Q.6 Find mean, median, mode, harmonic mean and Geometric mean for following data

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| Marks           | 0-10 | 10-20 | 20-30 | 30-40 | 40-50 | 50-60 |
|-----------------|------|-------|-------|-------|-------|-------|
| No. of Students | 2    | 5     | 8     | 16    | 9     | 5     |

Q.6

OR

- (a) Following data is regarding the number of letters in a word of a paragraph with 30 words:  
3,5,3,6,7,3,5,4,6,7,4,4,5,2,3,2,4,5,1,2,4,2,3,5,6,2,1,3,6,7

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(i) Form a discrete frequency distribution

(ii) Find mean

- (b) Find the missing frequency for the following frequency distribution. Given that  
Median = 24.

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| Class     | 0-10 | 10-20 | 20-30 | 30-40 | 40-50 |
|-----------|------|-------|-------|-------|-------|
| Frequency | 5    | 25    | ?     | 18    | 7     |

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— X —  
(3)

